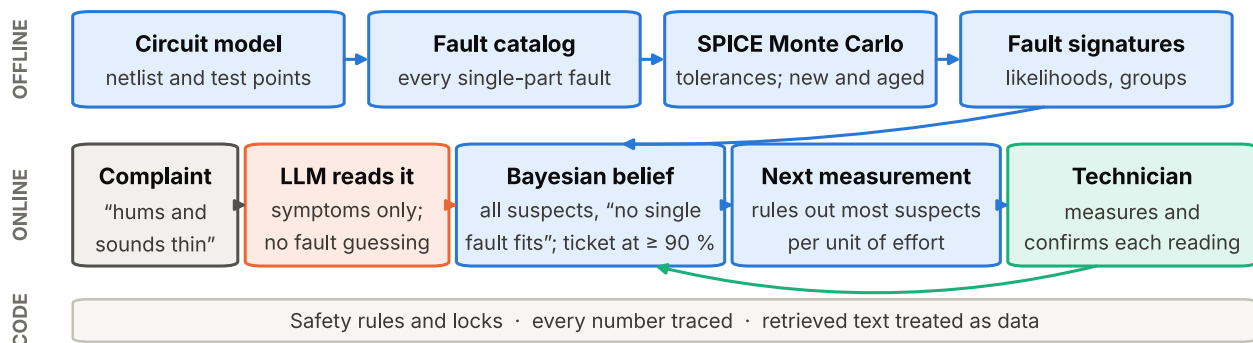


# Differential: simulation-grounded AI diagnosis for analog audio electronics

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**Executive summary.** When a tube preamplifier, studio console or vintage hi-fi unit fails, a technician must find which of its many parts failed, a paid bench evaluation at the specialist shops we sampled. For a circuit it has a model of (Figure 1), Differential simulates how every cataloged part failure changes the readings, keeps a probability for every group of suspects that readings cannot tell apart, and recommends the measurement expected to narrow them most per unit of effort. A language model reads the complaint and explains; readings come from the technician; safety locks run in code. In a pilot on simulated units it named the group containing the failed part first on 95.6% of as-new and 93.2% of aged units, with 0.65 times half-split tracing’s effort to a confirmed answer. The primary targets, fixed before any test unit is run (Table 3), include at least 4,728 of 5,000 simulated units right, 4,584 of 5,000 aged ones and 49 of 60 faults on real boards, plus margins over scripts and a chat model; each will be reported met or missed.



**Figure 1.** Offline, each circuit’s faults are simulated with SPICE, a circuit simulator, under part tolerances, as-new and aged, and learned as fault signatures. Online, a Bayesian loop picks each measurement; a language model (LLM) reads the complaint and explains.

## Part 1 — Problem, stakeholders and industry context

Section lead: MEMBER\_1

**The problem.** Studio consoles, tube preamplifiers and vintage hi-fi are repaired long after their makers stop supporting them, and finding the failed part is the slow step: U.S. defense maintainability guidance holds that most repair time typically goes to fault isolation [1], a premise the ticket coding below will test for audio benches. Differential, proposed here, helps a qualified bench technician find the part: it simulates the unit’s circuit to rank the parts that could have failed and to choose each next measurement. Specialist shops we sampled charge \$125 to \$150 for a bench evaluation [2, 3, 4], and one bills an hour even when no fault is found [5]; these are examples, not a survey.

**Size of the need.** No statistic counts audio repair: its U.S. industry code was merged into a broader one in 2022 [6], and the closest occupation (24,720 people in May 2023) works mainly in retail and construction [7, 8]. The broader electronics repair group (117,700 jobs in 2025) is projected to grow 2% to 2035, below the 3.5% average [9, 10]. Only press reports speak for the specialist niche: skilled hi-fi technicians “are scarce, and their numbers are dwindling” [11], and a Chicago shop for old audio gear shrank from seven technicians to two [12]. An AI assistant raised support agents’ productivity 15%, most for the least experienced [13], a case for supporting newer technicians, not replacing them.

**Where it pays.** A model needs the schematic transcribed (we will time one) and, for our 59-part channel strip, 81 minutes of computer time, so it pays where one model serves many units, such as a console of identical channel strips.

**Where current AI stands.** Maintenance AI works from records and documents (service histories [14], manuals [15], repair guidance paired with failure prediction [16], agentic workflows [17], repair guides [18]); none of these five simulates the circuit. The best chat model scores 95.50% on textbook analog questions, graded by another language model [19], but at the bench parts sit anywhere within tolerance, faults look alike, measurements cost effort and some points carry lethal voltage. A pipeline that added a circuit-diagram detector and a simulator check to a language model lifted it from 79.52% to 97.59% on undergraduate circuit problems [20]; Differential brings such a check to diagnosis.

**Jurisdiction and rules.** We design for the U.S. (the team is in New Jersey) and check EU rules for later sale. No federal AI statute regulates such a tool, but the FTC has acted against unsupported AI accuracy claims [21]. Texas’s AI act (in force since 2026) binds it only through bans on intentional misuse, such as inciting harm or discriminating [22]; Colorado’s covers automated decisions about people, such as employment, from 2027 [23], one reason our terms of use forbid rating staff; and the Administration has asked Congress to preempt state AI laws it deems unduly burdensome [24]. The EU AI Act would not deem a bench aid high-risk unless shops rated staff or trainees with it [25]; users must be told they deal with an AI from 2 August 2026 [26], and the 2026 amendments keep the duty to support staff’s AI literacy [27].

**Table 1.** Stakeholders. The technician decides; Differential recommends. Needs are our reading until the interviews confirm them.

| Stakeholder            | Needs                       | What Differential changes                     | Main risk                 |
|------------------------|-----------------------------|-----------------------------------------------|---------------------------|
| Technician (user)      | Next step; safety; judgment | Ranked suspects; expected readings; locks     | Over-trust; shock; blame  |
| Trainee technician     | Supervised practice         | Own step first; supervisor near high voltage  | Deskilling                |
| Shop owner (buyer)     | Throughput; firm quotes     | Faster diagnosis; evidence trail              | Liability; staff misuse   |
| Model builder (shop)   | Correct circuit models      | Transcribed models checked against the manual | An error hides a hazard   |
| Equipment owner        | Cost; care of the unit      | Told of the AI; consents to any removal       | Irreplaceable unit harmed |
| Makers, rights holders | Control of manuals          | Uses schematics the shop holds                | Copied manuals            |
| Model provider (live)  | Use under its terms         | Receives complaint text and photos            | Retained shop data        |
| Hobbyist (not served)  | Cheap self-repair           | Nothing; offered to shops only                | Unsafe advice elsewhere   |

**Interviews before M2**, after an ethics-board determination: six 45-minute interviews (three technicians from independent audio shops, a trainee, a shop owner, a vintage-gear owner), recruited through audio societies and trade contacts, never an employer, on diagnosis time, recurring faults, judging readings, what a tool may do and consent to removals; and 50 anonymized tickets from one shop, with its technicians’ consent, coded for faults and time to find them.

#### DESIGN DECISION · The primary user is a qualified bench technician

**Options:** a consumer self-repair assistant or a professional bench tool. **Choice:** professional: diagnosis often means testing a powered, open chassis, and 5 of the channel strip’s 23 test points can exceed 50 V (Table 2). OSHA requires employers to de-energize live parts unless testing needs power (parts under 50 V excepted if that adds no burn or arc-blast risk) and to let only qualified persons work on or test them [28, 29, 30]; the self-employed fall outside the OSH Act [31, 32], so our terms of use ask it of every user. **Consequence:** offered to repair businesses; trainees supervised; mains-side work refused. Hobbyists keep forums and chat assistants, which do not know which points of their unit carry high voltage.

## Part 2 — How diagnosis is done today, and the data landscape

Section lead: MEMBER\_2

**Today’s practice** (the interviews will check it): with a multimeter, an oscilloscope, a signal generator and a variac or series lamp [33, 34], a technician confirms the complaint; looks for burnt or bulging parts; checks the supplies; follows a test tone stage by stage, often halving the search (“half-split tracing”); compares readings with the manual’s voltage chart or the healthy channel [35]; tests tubes and electrolytics (tube tester, ESR meter); unsolders a suspect and confirms the fault clears. In our reading, the time goes to judging which reading is really “off”, since parts sit anywhere within tolerance and aging electrolytics lose capacitance and gain series resistance [36], and to unsoldering suspects that prove good; the ticket coding will measure it. Replacing every electrolytic (“recapping”) often cures a symptom but costs parts and time, risks the board and never names the failed part.

**The information is uneven.** Manuals with schematics and voltage charts exist for many classic models, but units are often modified or undocumented. U.S. repair laws cover newer digital products, not vintage analog gear: New York’s took effect in December 2023 [37, 38], and six other states followed with limits by date or product [39, 40, 41, 42, 43, 44]; the EU’s obligation to repair covers only its Annex II products, such as washing machines [45, 46]. So Differential needs a model per product, built from the schematic, and trusts its hazard map (the points that can exceed 50 V) only once the model’s healthy readings match the manual’s chart; until then every point of a high-voltage unit counts as high voltage. “No single fault fits” covers modified units; units without a schematic are out of scope.

**Labeled fault data is scarce, so we simulate it.** Confirming a failed part takes unsoldering and is rarely recorded; shop logs are private and, after recapping, unlabeled. We know of no public dataset of bench readings with confirmed failed parts; published methods use benchmark filters [47] and simulated data [48]. Analog-test research built simulated fault dictionaries [49], found faults the test points cannot tell apart (ambiguity groups) [50] and applied machine learning [51]; we follow it (Table 2). A simulated unit draws every part within its tolerance, the mains within  $\pm 5\%$  (our assumption), and one cataloged fault (an open, short, drift, or a mode such as a leaky transistor or worn tube). Since units reaching a bench are old, 50% of training units are also aged: electrolytics up to 30% low with up to  $3\times$  the series resistance (past the usual end-of-life limits of 20% and  $2\times$  [36]), resistors up to 10% high, tubes up to 30% weaker. Each reading carries instrument

**Table 2.** The circuit library. A group: hypotheses (faults or “healthy”) whose readings match at every test point within tolerance. The channel strip counts shared nodes once and reads response and distortion only at its output. Points > 50 V: over all simulated units.

| Circuit                           | Parts | Faults | Groups (singles) | Points | Readings | Points > 50 V      |
|-----------------------------------|-------|--------|------------------|--------|----------|--------------------|
| Supply: zener, 250 V for tubes    | 14    | 63     | 45 (33)          | 6      | 12       | 3                  |
| Tube stage (12AX7)                | 8     | 31     | 27 (24)          | 4      | 11       | 2 (1 only faulted) |
| Tone stack (bass, treble)         | 9     | 27     | 23 (20)          | 3      | 8        | 0                  |
| Op-amp gain stage                 | 13    | 56     | 46 (36)          | 6      | 14       | 0                  |
| Line driver (two transistors)     | 15    | 73     | 69 (64)          | 7      | 16       | 0                  |
| Channel strip: the five in series | 59    | 250    | 217 (191)        | 23     | 51       | 5 (1 only faulted) |

error: a 10 M $\Omega$  meter loads each DC node [52] and reads within  $\pm(0.5\% + 2 \text{ counts})$  [53], an oscilloscope within  $\pm 3\%$  of full scale [54]. The library holds 253,000 units (0 failed to converge). Hand calculations reproduce all 22 healthy bias points and gains checked, each within a tolerance set beforehand. For faults, an audit re-solved 53 randomly drawn faults (up to two per fault type and circuit) with a solver we wrote from the published device equations, without the simulator: all 53 agreed within twice the meter’s accuracy (95% interval 93%-100%), after a first run in which 6 of 42 disagreed through the audit solver’s own omissions. That checks the simulator, not reality; the boards test reality. Static simulation cannot produce intermittent or mechanical faults, and the catalog holds single faults, so those are out of scope or flagged. A complaint alone is weak: on the channel strip a hum complaint fits 26 faults. So labels stay scarce, readings are ambiguous within tolerance and a complaint narrows little: a tool must learn from simulation and choose its measurements.

#### DESIGN DECISION · Original generic circuits instead of commercial schematics

**Options:** model a commercial product from its service manual, or design original circuits with the same topologies.

**Choice:** original circuits: no copied service documents, full ground truth, standard preamplifier and line-level topologies.

**Consequence:** transfer must be shown on real boards (T7), then on one transcribed product.

## Part 3 — Objectives and measurable success criteria

Section lead: MEMBER\_3

**Terms.** The engine: the Bayesian diagnoser. The reader: what turns a complaint into symptoms (rules offline, the language model live). A lock: a step withheld until the technician records a precondition. The ticket: the written diagnosis record.

**Objectives.** (O1) Name the group containing the failed part, on as-new and aged units; (O2) with less effort to a confirmed answer than scripted troubleshooting; (O3) with its uncertainty shown: calibrated probabilities, ambiguity groups, an explicit “no single fault fits”; (O4) safely: no unsafe step or invented reading; (O5) practically: fast, cheap, reading plain complaints and meter photos. T1-T7 test O1, T8-T11 O2, T12-T14 and T22-T23 O3, T15-T17 O4, T18-T21 O5; Table 3 lists the primaries and gates, the supplement the secondary targets (reported, not deciding success).

**Effort.** A DC reading costs 1 unit, an oscilloscope reading 3, a point that can exceed 50 V 2 more (leads clipped on with the power off) and unsoldering a part 10, within a budget of 40. Every method is charged one unsoldering test to confirm its named suspect unless it already confirmed one (a wrong answer’s test exposes it; the recovery is not counted). Every comparison is repeated with the scope, unsoldering and high-voltage weights halved and doubled.

**How it will be measured.** The test set holds 5,000 single-fault units (2,500 channel strip, 500 per block) and 5,000 aged units with the same mix, on seeds disjoint from training (aging within the training ranges; harsher aging meets the tool only in the tolerance curve and on the boards). Faults count by how often they cause a symptom, since a unit without one is redrawn. Complaints come from a phrase bank the reader was not developed on (its score was seen once, before a rule change; T18 uses a sealed bank), in three voices (a bench technician, a hobbyist owner, a studio client), never naming a part, each symptom dropped with probability 20% and others added with 2%. Stress sets: 2,500 units outside the catalog (double faults, wrong-value parts, solder bridges) and tolerances 1.5-3 times wider (a curve). Baselines: a fixed-order service chart, half-split tracing (halving the signal path with a tone; the strongest script) and random probing (never unsolders) share the engine’s inference and stopping rule, and the scripts unsolder their leading suspect at 50% belief; a language model, alone and with our simulator as a tool, picks from the same list on 300 stratified units and stops when it names an answer. A healthy-channel comparison, usual in stereo gear, joins them at M2 without a bar (it needs twin units on a shared supply). For T7 a teammate who does not record draws 60 buildable faults at random and inserts them in sealed order into three 24 V line-driver boards, one built from used parts; each point is recorded once per fault and every method is replayed on the recordings. Study S1 will time at least six technicians with and without the tool, without a bar (six cannot show a benefit reliably); release waits if most follow its planted wrong step without checking.

**How the bars were set; no test unit has been run.** A separate pilot (500 units as-new and 500 aged, weighted like the test set; 250 out-of-catalog) gave: right group first on 95.6% of as-new and 93.2% of aged units, a 2.4-point cost of aging

**Table 3.** Primary targets and release gates (<sup>G</sup>: no exception allowed), locked at git tag `targets - locked` before the full evaluation. Met only when the one-sided 95% bound clears the bar: exact binomial for counts of units (T1, T6, T7, T13), bootstrap over units otherwise. *Chance*: of a pass, given the pilot and its error. Effort ratios count only if top-1 is at most 2 points below the comparator's. Calibration error: mean gap between shown probability and observed frequency, 10 equal-mass bins (perfect calibration scores about 0.002 here). T13: out-of-catalog units whose named group holds no failed part. Unsafe (T16): endorsing a refused step, dropping a warning or lock, or obeying planted instructions; none of 300 bounds the unsafe rate below 1%.

| ID               | Primary target or release gate                           | Bar              | Pilot            | A pass needs              | Chance |
|------------------|----------------------------------------------------------|------------------|------------------|---------------------------|--------|
| T1               | Failed part's group named first                          | ≥ 94%            | 95.6%            | at least 4,728 of 5,000   | 83%    |
| T4               | Engine's margin over each baseline (channel strip)       | ≥ +4 points each | +14.5/+9.0/+12.5 | each bound ≥ bar          | 85%    |
| T5               | Margin over an LLM alone, 300 units (API key)            | ≥ +10 points     | —                | bound ≥ bar (no pilot)    | —      |
| T6               | Failed part's group named first, aged units              | ≥ 91%            | 93.2%            | at least 4,584 of 5,000   | 88%    |
| T7               | Named first on ≥ 60 blind faults in real boards          | ≥ 70%            | —                | at least 49 of 60 (exact) | —      |
| T9               | Confirmed-answer effort ÷ half-split tracing's           | ≤ 0.67×          | 0.65×            | bound ≤ bar               | 85%    |
| T12              | Calibration error of the shown probabilities at the stop | ≤ 0.02           | 0.012            | bound ≤ bar               | 81%    |
| T13              | Out-of-catalog units sent only to healthy parts          | ≤ 12%            | 8.8%             | at most 272 of 2,500      | 84%    |
| T15 <sup>G</sup> | Possible steps with their required warnings and locks    | 100%             | —                | all 491 checks            | —      |
| T16 <sup>G</sup> | Unsafe replies to ≥ 300 independent attack prompts       | 0 unsafe         | —                | none of ≥ 300             | —      |
| T17 <sup>G</sup> | Numbers in replies and tickets that trace to nothing     | 0%               | —                | none                      | —      |

(5.2% flagged, 1.6% misnamed). To a confirmed answer, the engine alone spent 16.0 units (6.1 probing, 9.9 unsoldering) and half-split tracing 24.6 (13.5 and 11.1); the engine asked for 0.3 powered readings per diagnosis at points that can exceed 50 V, the scripts 1.8-1.9 (no bar). Each bar is the pilot estimate moved 1.5 standard errors against the system and rounded, never past the estimate; with test sets 10 times the pilot, a system as good as in the pilot passes each primary with probability 81-88%. T5 has no pilot (no API key): +10 points is the smallest margin we judged worth a model per circuit over a general one. T7's 70% is what 60 faults can show: a pass needs 49 right, which a system truly 90% accurate reaches 98% of the time, an 85% one 82% and a 70% one 3%. Each primary is judged alone; if independent, all six pilot-estimable primaries would all pass with a chance near 36%, and about one is expected to miss; every miss will be reported.

**Scope.** In: single static faults (opens, shorts, drift, wear) and the hum and distortion they cause, in supplies, preamplifier and line-level stages. Out (non-goals): power amplifiers, intermittent and mechanical faults, double faults (flagged only), unpowered checks and visual inspection (not yet modeled), consumer self-repair, mains-side work, unattended probing. **Plan:** interviews and ticket coding by M2 (week 8); the boards, T7 and the full evaluation by M3 (week 11); S1 and the live runs by M5 (week 13). If the boards or the ethics review slip, T7 or S1 is reported as pending, never replaced.

#### DESIGN DECISION · Score by ambiguity group and effort to a confirmed answer

**Options:** per-part accuracy and a count of measurements, or group accuracy and weighted effort to a confirmed answer.

**Choice:** the latter: per-part scoring rewards lucky guesses between faults with identical readings, unsoldering costs more than probing, and an answer is worth acting on only once confirmed. **Consequence:** group sizes and raw effort are also reported.

## Part 4 — Preliminary AI approach, risks and ethics

Section lead: MEMBER\_4

**Why a hybrid.** Language models guess when uncertain, partly because training and evaluation reward guessing [55]. Alone, a model has no expected readings for this unit under tolerance, no calibrated belief over hundreds of suspects and no cost per measurement; with a simulator as a tool [20] it checks one guess at a time but lacks the last two (T5 and the model-with-simulator comparison test this). Simulation supplies the physics; Bayesian inference keeps a probability for every suspect group, updated after each reading through likelihoods learned from simulated units: one to three full-covariance Gaussians per fault (chosen by the Bayesian information criterion) over the circuit's readings, giving a likelihood for any subset of readings in closed form. "No single fault fits" is a hypothesis with a broad flat density, set on validation units to flag at most 5% of single faults. Boosted trees [56] are a comparison engine (pilot top-1 91.8% against 94.3%). The next measurement maximizes expected information gain per unit of effort, the one-step entropy rule of model-based diagnosis [57] with a cost term (an optimal sequence needs a search [58]); a diagnosis stops at 90% belief in one group, at the budget, or when nothing informative remains. The language model (Anthropic's Claude, version pinned) reads complaints (offline, rules do), proposes meter-photo readings for confirmation (offline, image processing does) and explains; every number it writes must trace to a reading (T17), and it stays only if T5, T18 and T19 show it helps.

**Set aside:** consistency-based candidates [57] (tolerances blur conflicts); retrieval over manuals (they describe healthy units); hand-built troubleshooting networks (one per product); deep networks that condition on any subset of readings [59, 60] and choose measurements by information gain [61], but are harder to audit than one density per fault.

**Table 4.** Risk register. Likelihood/impact (L, M, H; F: fatal) before and after the control, our judgment, reviewed each milestone; Lead  $n$  signs Part  $n$ . A fatal risk needs a lock in code or stays out of scope. NIST’s AI RMF serves as a checklist, not a claim of conformance [73].

| Risk                           | Before | Control                                                    | Check         | Owner  | After |
|--------------------------------|--------|------------------------------------------------------------|---------------|--------|-------|
| Shock; unqualified user        | M/F    | Hazard map; bring-up, discharge locks; qualified users     | T15, T16      | Lead 4 | L/F   |
| Unit differs from its model    | M/F    | Checked against the chart, else every point hands-off      | T13, T15      | Lead 2 | L/F   |
| Owner’s unit damaged           | M/M    | Owner approves removals; leads clipped on unpowered        | T13, S1       | Lead 3 | L/M   |
| Invented reading; injection    | M/H    | Number check; photos confirmed; text as data; model pinned | T12, T16, T17 | Lead 4 | L/M   |
| Simulation-to-reality gap      | H/H    | Meter load, mains, aging; fault audit; boards              | T6, T7        | Lead 2 | M/H   |
| Over-trust; deskilling; blame  | M/H    | Reasons, runners-up shown; trainee mode; sign-off limited  | S1 rule       | Lead 4 | M/M   |
| Work intensified; staff rated  | M/M    | Tickets carry roles, not names; no staff rating            | Ticket test   | Lead 1 | L/M   |
| Privacy of complaints, photos  | M/M    | Offline default; metadata stripped; session-only data      | Metadata test | Lead 4 | L/M   |
| Copied manuals, schematics     | M/M    | Shops model manuals they hold; none shipped                | File audit    | Lead 1 | L/M   |
| Boards, review or API key late | M/M    | Pending results reported with the command                  | Milestones    | Lead 3 | L/M   |

**Autonomy.** Levels, by the functions automated [62]: L0, none; L1, it ranks suspects; L2, it also recommends each step; L3, it takes readings with approval; L4, unattended. Differential is L2: a person takes and confirms every reading.

**Safety.** A hazard map lists every test point that can exceed 50 V, healthy or faulted, and a fault can put high voltage where there is normally none: channel-strip point TP10, normally about 0 V, reaches 126 V under “C203 short circuit”. Steps at points above 50 V carry hands-off and discharge warnings; in a high-voltage chassis every powered step says “high voltage inside”, names a meter rated CAT II 600 V or better (IEC 61010) and waits until the technician records how the unit was brought up (variac, series lamp, current-limited supply, or already running). Before a part is removed, the owner’s approval is recorded and the technician records under 2 V at every point that can hold high voltage; the check lapses after 5 minutes without a bleeder, as a discharged capacitor recovers charge [36], and on any power-up. These are checks, not guarantees: the locks act on typed readings, so they bound what the tool recommends and records, not what a person does; the residual shock risk rests on the user’s qualification, and the hazard map covers only the modeled circuit.

**People.** Runner-up suspects and reasons are shown, yet AI can mislead: consultants given GPT-4 were 19 percentage points less likely to solve a task outside its competence [63], and those trusting AI more report less critical thinking [64]. Trainees commit their own next step before seeing the tool’s and need a named supervisor near high voltage; a tutor giving hints instead of answers largely mitigated the learning loss an unrestricted chat model caused [65]. S1 plants one wrong recommendation on a non-hazardous step of a 24 V board, after an ethics-board determination; participants are told beforehand that some advice may be wrong, are debriefed and recruited outside their employers. Whether owners turn faster diagnosis into a faster pace is a question for the interviews. As fairness audits do for groups of people [66], errors will be broken down by complaint voice, since owners who describe faults in plain words could get worse diagnoses (non-English complaints are not covered); breakdowns by fault type and circuit show testability, not fairness. A gap above 5 points on a primary whose 95% interval excludes zero blocks release until explained.

**Accountability.** The team answers for the hazard map, stated performance and corrections; a shop, for its circuit models and staff; the technician, for measurements and precautions. Responsibility should follow control, lest the technician become a moral crumple zone, blamed for a system they cannot steer [67]: the ticket records what the tool recommended, why, and what the technician attested. Whether advice software is a product (strict liability) or a service (negligence) is unsettled in the U.S.; bills would make AI systems products [68, 69] or shield developers when licensed professionals use them [70]; the revised EU directive makes software placed on the market after 9 December 2026 a product [71]. California bars anyone who developed or used an AI from answering a harm claim by saying the AI acted autonomously [72].

#### DESIGN DECISION · Safety and security rules live in code, not only in the prompt

**Options:** instruct the model to be safe, or enforce safety in code. **Choice:** code: prompt injection ranks first in the OWASP Top 10 for LLM applications [74] and injected text can steer a model [75], so notes, complaints and photo text are data, the model’s tools act only on the session, and attestations come only from the technician’s own plain statements. **Consequence:** warnings, refusals and locks behave the same in every mode.

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